

Is the 25 mK anomaly of $\text{Ce}_2\text{Hf}_2\text{O}_7$ the emergent photon?

**A torus-artifact-free exact-diagonalization test and a
parameter-free entropy budget**

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Abstract

The heat capacity of $\text{Ce}_2\text{Hf}_2\text{O}_7$ shows two peaks, at $T_1 \simeq 64.6$ mK and $T_2 \simeq 24.2$ mK, and the lower one has been discussed as the crossover into the quantum-spin-ice gauge sector. We test that identification with exact diagonalization of the nearest-neighbour dipole–octupole XYZ model, after removing a finite-size effect that has not previously been separated from the physics. A periodic cluster is a torus and supports ice-preserving spin rearrangements that close only through a periodic image; they enter at order $J_{\pm}^2/J_{zz}$, one order *before* the physical hexagon ring exchange at $12|J_{\pm}|^3/J_{zz}^2$, and they manufacture the dominant low-temperature scale. At the published NLC-A couplings this artifact places a spurious heat-capacity peak at 28.7 mK — within 19% of the measured T_2 , and therefore capable of confirming the gauge interpretation for the wrong reason. Deleting the artifact exactly, by masking transport-changing matrix elements of the Feshbach map, moves the nearest-neighbour gauge crossover to 6.0 mK. Across 66 certified points of the fit-constrained coupling region the largest attainable ratio $T_{\text{gauge}}/T_{\text{spinon}}$ is 0.198, against the measured 0.374. We supplement this model-dependent statement with a model-independent one: any crossover that is the ordering of the ice manifold must release the Pauling entropy $\frac{R}{2} \ln \frac{3}{2} = 1.687 \text{ J mol}^{-1} \text{ K}^{-1}$, i.e. 29% of $R \ln 2$, and any gauge-sector crossover must be followed below it by a photon T^3 tail. We show what the currently available digitized data can and cannot decide, and specify the measurement — heat capacity below 20 mK — that settles it. Our conclusion is that within the nearest-neighbour XYZ model the 25 mK feature is not the gauge crossover; the honest alternatives are beyond-nearest-neighbour exchange, a larger $|J_{\pm}|$ than the high-temperature fit constrains, or a non-gauge origin.

I. INTRODUCTION

$\text{Ce}_2\text{Hf}_2\text{O}_7$ is one of a small set of dipole–octupole pyrochlores in which a quantum-spin-ice (QSI) ground state is plausible on symmetry, chemistry and spectroscopic grounds [1–5]. The two-peak heat capacity reported by Smith *et al.* [6] is the sharpest quantitative handle available: an upper feature near $T_1 \simeq 65$ mK, naturally read as the spinon (magnetic monopole) crossover, and a lower feature near $T_2 \simeq 25$ mK for which the emergent photon is

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the natural candidate. If correct, this is the first thermodynamic sighting of the QSI gauge sector.

The claim is calibrated by numerics, and the numerics have a problem that is topological rather than statistical. Every finite periodic cluster of a pyrochlore is a three-torus. On a torus there exist ice-preserving spin rearrangements whose flipped set closes only after a periodic identification. Such a process does not exist on the infinite lattice, and on the clusters used in practice it is *shorter* than the physical process it competes with. Its amplitude and that of the hexagon ring exchange are

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}, \quad g_6 = \frac{12|J_{\pm}|^3}{J_{zz}^2}, \quad \frac{g_4}{g_6} = \frac{J_{zz}}{3|J_{\pm}|}, \quad (1)$$

so the artifact does not shift the gauge scale, it *replaces* it, with a different power of the coupling. At the couplings relevant to $\text{Ce}_2\text{Hf}_2\text{O}_7$, $|J_{\pm}|/J_{zz} \simeq 0.125$, the ratio is $g_4/g_6 \simeq 2.7$ and the artifact wins.

This paper does three things. Section III shows that at the published NLC-A couplings the artifact peak lands at 28.7 mK — on top of the measured 24.2 mK anomaly — so a small-cluster calculation appears to confirm the photon interpretation while measuring the box. Section IV removes the artifact exactly and reports what the nearest-neighbour model then predicts. Section V steps outside the model entirely and gives two parameter-free constraints, an entropy budget and a T^3 tail, that the data must satisfy for *any* gauge interpretation. Section VI states the limitations honestly; they are substantial and they all push in the same direction.

II. MODEL, COUPLINGS, AND THE DIGITIZED DATA

We use the nearest-neighbour pseudospin Hamiltonian of Eq. (2) below in the exchange-diagonal local frame,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + \text{h.c.}) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + \text{h.c.}), \quad (2)$$

related to the published (J_a, J_b, J_c) parametrization by

$$J_{zz} = J_a, \quad J_{\pm} = -(J_b + J_c)/4, \quad J_{\pm\pm} = (J_b - J_c)/4. \quad (3)$$

The three published seeds of Ref. [6] are, in meV,

$$\begin{aligned} \text{NLC-A} &= (0.050, 0.021, 0.004), \\ \text{NLC-B} &= (0.051, 0.008, -0.018), \\ \text{QMC (ordered)} &= (0.046, -0.003, -0.010). \end{aligned} \tag{4}$$

NLC-A maps to $(J_{\pm}, J_{\pm\pm})/J_{zz} = (-0.125, +0.085)$. Note that $J_{\pm\pm}$ is not small: omitting it is not a controlled approximation for this material, so all calculations below use the full XYZ Hamiltonian and its complete 2^{16} cubic-16 basis where required.

Measured peak positions, extracted from the published figure, are $T_2 = 24.19$ mK and $T_1 = 64.65$ mK, giving

$$\mathcal{R}_{\text{exp}} \equiv T_2/T_1 = 0.374. \tag{5}$$

The digitized trace begins at $T = 20.6$ mK with $C_{\text{mag}} = 1.93 \text{ J mol}_{\text{Ce}}^{-1} \text{K}^{-1}$, rises to 2.25 at 23.7 mK and falls thereafter. **There is no digitized data below the low-temperature peak**, a fact that limits Sec. V and that we do not paper over. The extraction carries no experimental uncertainties; it is used for visual comparison and for the integrals of Sec. V, not for a χ^2 .

III. THE ARTIFACT LANDS ON THE ANOMALY

Table I compares, at fixed couplings and on the same cluster and the same virtual space, the low-temperature heat-capacity peak of the naive periodic calculation with the transport-masked one of Ref. [7]. The masking deletes every ice-block matrix element of the exact Feshbach map $F(z) = PHP + PHQ(z - QHQ)^{-1}QHP$ that connects different transport (polarization) sectors; a contractible rearrangement preserves the label, a rearrangement closed through a periodic image changes it by a box vector. No counterterm and no fitted coefficient is involved.

A second, independent extraction on the same cluster with the full two-peak analysis gives, at NLC-A, a periodic low peak at 28.7 mK against a masked one at 6.0 mK, and at $(-0.152, 0.085)$ a periodic 37.5 mK against a masked 8.1 mK. The two extractions differ in how the peak is located in a multi-feature curve; the qualitative statement is identical and is the point of this section:

TABLE I. Low-temperature heat-capacity peak at the $\text{Ce}_2\text{Hf}_2\text{O}_7$ -relevant couplings, in mK, using $J_{zz} = J_a$ of the corresponding seed. “periodic” is naive exact diagonalization of the same cluster and virtual space; “masked” has the torus artifact deleted. The measured anomaly is at 24.2 mK.

$(J_{\pm}, J_{\pm\pm})/J_{zz}$	periodic (mK)	masked (mK)	masked/ g_6
(−0.050, 0.000)	7.52	0.71	0.821
(−0.125, 0.085) (NLC-A)	17.40	6.01	0.442
(−0.150, 0.000)	17.40	7.72	0.329
(−0.152, 0.085)	17.40	8.11	0.335

At the published couplings, naive periodic exact diagonalization predicts a low-temperature heat-capacity peak within 19% of the measured 24.2 mK anomaly. Essentially all of that peak is the torus artifact. Removing it moves the prediction down by a factor of 3–5.

The coincidence is not a small perturbation of a correct answer. Because $g_4 \propto J_{\pm}^2$ and $g_6 \propto |J_{\pm}|^3$, the artifact carries a different power of the coupling: a log–log fit of the peak position against $|J_{\pm}|$ gives 1.80 for the periodic branch and 2.79 for the masked branch, approaching the second and third orders of Eq. (1) with no fitted subtraction. A calculation that reproduces T_2 through the artifact would also reproduce the wrong dependence on pressure, chemical substitution, or any other knob that moves $J_{\pm}/J_{zz}$.

The same artifact has a second, unrelated symptom worth recording for anyone calibrating on small clusters: it inverts the measured gauge flux. The winding amplitude g_4 is even in $J_{\pm}$ while g_6 is odd, so for $J_{\pm} < 0$ — the sign relevant to every Ce pyrochlore — artifact and physics oppose and the artifact wins. On the 16-site cluster the raw ground state reports $\langle O_{\text{hex}} \rangle > 0$ (0-flux) over the entire (J_x, J_y) plane, whereas the winding-free state reports π -flux throughout $J_{\pm} < 0$.

IV. WHAT THE NEAREST-NEIGHBOUR MODEL PREDICTS ONCE THE ARTIFACT IS GONE

A. The peak-ratio gate

The absolute position of a finite-cluster gauge peak carries a host-dependent prefactor: the combinatorial hexagon-flippability coefficient is 1.070 on the 16-site cubic cluster and 0.521 on the 32-site FCC cluster, so the prediction moves down by roughly a factor of two on the larger host and is not converged even there. The *ratio* of the two peaks is a better-controlled observable, because the spinon peak sits at $\simeq 0.25 J_{zz}$ and is untouched by the mask (we verify it moves by less than 1% for $|J_{\pm}|/J_{zz} \leq 0.09$).

We therefore scan the fit-constrained region and ask whether any certified point reaches $\mathcal{R}_{\text{exp}} = 0.374$. Of 80 grid points, 66 are certified in the sense that the complete ice-descended band is resolved (all 90 roots on cubic-16); the remainder dissolve into the spinon continuum and cannot support any peak statement at all. The result is

$$\max_{\text{certified}} \mathcal{R}_{\text{masked}} = 0.198 \quad \text{at} \quad (J_{\pm}, J_{\pm\pm})/J_{zz} = (+0.050, +0.085), \quad (6)$$

with the next two values 0.121 and 0.111. The maximum sits at *positive* $J_{\pm}$, i.e. in the 0-flux sector, and is still short of the measurement by a factor 1.9. Restricted to the physically favoured π -flux side the shortfall is far larger.

[PENDING: The repository contains a second, more conservative estimate of the same quantity — “largest reachable ratio 0.050 at $J_{\pm}/J_{zz} = -0.17$, shortfall 7.5 $\times$ ” — computed with a different peak-identification rule. The two must be reconciled, and one rule adopted, before submission. Both support the null; they differ by a factor of four in how badly.]

B. Best available fit

Ranking the winding-free curves against the digitized C_{mag} over 0.025–2.35 K by unweighted RMSE in $\log T$ selects $(J_{\pm}, J_{\pm\pm})/J_{zz} = (-0.156, +0.115)$ with $J_{zz} = 0.236$ K = 0.0203 meV. That point puts its two winding-free maxima at 3.53 mK and 67.2 mK. The high peak is essentially on the measured T_1 ; the low peak is 6.9 times below the measured

T_2 . The nearest-neighbour model can reproduce the spinon scale and the high-temperature tail while missing the low peak by almost an order of magnitude.

This is consistent with an independent observation about the fit itself. At high temperature the heat capacity depends on the exchange parameters only through $\sum J^2$ at leading order, so raising $|J_{\pm}|$ while lowering J_a to hold $\sum J^2$ fixed is a nearly flat direction of the high-temperature objective. An order-5 exact numerical linked-cluster calculation reproduces the published 0.25–1.2 K data with mean residual 2.2% at a point with $(J_{\pm}, J_{\pm\pm})/J_a = (-0.156, 0.115)$, better than either published minimum (3.7% for A, 5.1% for B). **The published fit does not constrain the gauge scale**, because the gauge scale is set by $|J_{\pm}|^3$ and lives an order of magnitude below the fitted window. The material is, on this evidence, more quantum than the fit says — but not by enough to close the factor of seven.

V. TWO CONSTRAINTS THAT DO NOT DEPEND ON THE MODEL

Because the model-dependent test above inherits a finite-size prefactor, we add two constraints that any gauge interpretation must satisfy regardless of Hamiltonian.

A. The Pauling entropy budget

If the lower peak is the ordering of the emergent gauge field within the ice manifold, then integrating the magnetic heat capacity across it must release exactly the residual ice entropy [8, 9],

$$S_{\text{ice}} = \frac{R}{2} \ln \frac{3}{2} = 1.687 \text{ J mol}_{\text{Ce}}^{-1} \text{K}^{-1} = 0.293 R \ln 2. \quad (7)$$

Equivalently, of the total $R \ln 2 = 5.763 \text{ J mol}^{-1} \text{K}^{-1}$ available to an effective spin-1/2 doublet, 71% must be released above the ice plateau (by the spinon peak and the high-temperature tail) and 29% below it. This is a counting statement about the ice manifold, not about any effective gauge Hamiltonian, and it holds for the π -flux and 0-flux sectors alike.

The test is then:

$$\Delta S_{\text{low}} = \int_0^{T_{\text{dip}}} \frac{C_{\text{mag}}(T)}{T} dT \stackrel{?}{=} S_{\text{ice}}, \quad (8)$$

with T_{dip} the minimum between the two peaks. A result significantly below S_{ice} means the low peak is not the ice-manifold crossover (something else, e.g. a fraction of sites or a disorder-broadened subset, is ordering); a result significantly above it means the peak is not confined to the ice manifold.

[PENDING: Eq. (8) has not yet been evaluated. The obstruction is real: the digitized trace starts at 20.6 mK, above the peak's low-temperature flank, and the unmeasured tail is not negligible — with $C_{\text{mag}}(20.6 \text{ mK}) = 1.93$, a T^3 tail contributes $\simeq 0.64$ and a linear tail $\simeq 1.93 \text{ J mol}^{-1}\text{K}^{-1}$, i.e. between 38% and 114% of S_{ice} by itself. The partial integral from 20.6 mK to the dip is computable now and is worth reporting with the tail treated as a stated systematic; a decisive test requires author data below 20 mK. Script: analysis/entropy_budget.py.]

B. The photon T^3 tail

Below a genuine gauge crossover the specific heat of a linearly dispersing emergent photon with two polarizations must be

$$\frac{C}{N} = \frac{2\pi^2}{15} \left(\frac{k_B T}{\hbar c} \right)^3 \frac{V}{N}, \quad (9)$$

with c the emergent light speed [10]. This is the most distinctive fingerprint the gauge sector has, and it is a two-parameter-free prediction once c is known: the exponent is 3 and the coefficient is fixed by c , which is itself computable from the ring-exchange amplitude. A measurement of C_{mag} from 5 to 20 mK therefore does more than fill in a tail — it discriminates the photon (T^3) from a Schottky anomaly (activated), a nuclear contribution (T^{-2}), a spin-glass or disorder contribution (roughly linear), and a conventional magnon (T^3 but with a coefficient set by the ordered moment and a magnetic Bragg peak to accompany it). We regard this, not the peak position, as the decisive measurement.

VI. LIMITATIONS, AND WHICH WAY THEY PUSH

1. **Cluster size.** The coupling-plane scan is on the 16-site cluster, the only one on which the complete $J_{\pm\pm}$ basis is affordable. The host coefficient falls from 1.070 to 0.521 between that cluster and the 32-site FCC cluster, i.e. the larger and more reliable

host pushes the gauge peak *colder*, widening the discrepancy rather than closing it. This is the single most important caveat and it does not rescue the gauge reading.

2. **Virtual-depth truncation.** On the larger host the virtual space must be truncated; the measured error of the shallowest truncation is up to 40% low at strong coupling, again in the direction of a colder peak.
3. **Band completeness.** For $J_{\pm}/J_{zz} \lesssim -0.19$ the ice-descended band is no longer isolated from the spinon continuum on this cluster and no peak statement is admissible; the minimum ice-manifold overlap falls from 0.89 at $J_{\pm}/J_{zz} = -0.05$ to 0.36 at -0.18 . The scan is gated on this, which is why 14 of 80 grid points are excluded.
4. **Digitized data.** No experimental uncertainties are available, so we quote no χ^2 and no confidence interval, and the entropy integral is incomplete below 20.6 mK.
5. **Nearest-neighbour only.** The model contains no further-neighbour or dipolar exchange. The experimental paper itself raises beyond-NN terms as a possibility, and our result is one of the reasons to take that seriously.

Every numerical caveat pushes the predicted gauge peak to lower temperature. The null result is therefore robust in the only direction that matters.

VII. CONCLUSION

Within the nearest-neighbour dipole–octupole XYZ model, and after the torus artifact is removed exactly, no certified set of couplings in the fit-constrained region places a gauge-sector heat-capacity feature at 25 mK relative to a 65 mK spinon feature: the best attainable ratio is 0.198 against a measured 0.374, and the best RMSE fit misses the low peak by a factor of seven. The coincidence that makes this worth stating carefully is that the *uncorrected* periodic calculation does place a peak at 28.7 mK, so a small-cluster calibration confirms the gauge interpretation by reproducing its own finite-size artifact.

We do not claim that $\text{Ce}_2\text{Hf}_2\text{O}_7$ is not a quantum spin ice; the spinon scale and the high-temperature thermodynamics are reproduced well, and the flat direction of the high-temperature fit means the true $|J_{\pm}|$ may be appreciably larger than published. We claim

that the 25 mK feature is not accounted for as dressed gauge dynamics of the nearest-neighbour model, and that the question is settled not by a better fit but by a measurement: the heat capacity between 5 and 20 mK, whose exponent and entropy content decide the matter without reference to any Hamiltonian.

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Heat-capacity data digitized from Ref. [6]; the winding-free construction is that of Ref. [7].

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